

Modern Physics Notes

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Abstract

This is a simple abstract for the Modern Physics notes.

1 Introduction

Class test:

10% $\times$ 2 (special relativity + quantum mechanics)

+ 80% exam (4 short (2+2) + 2 long)

2 Galileo Relativity

2.1 reference frame

It's about Newton's first law: An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction, unless acted upon by a net external force. And clearly, a reference frame that moves relatively to an inertial reference frame at a constant speed is also an inertial reference frame.

- Inertial reference frame: a reference frame in which Newton's first law holds.
- Non-inertial reference frame: a reference frame in which Newton's first law does not hold.

It is really important because when we use different reference frames to describe the motion of the same object, we may get different results.

2.2 velocity and acceleration

Velocity is the rate of change of position with respect to time, and acceleration is the rate of change of velocity with respect to time.

$$\begin{aligned} \text{velocity} \quad \vec{v} &= \frac{d\vec{r}}{dt} \\ \text{acceleration} \quad \vec{a} &= \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} \end{aligned}$$

For different theories, the definitions of velocity and acceleration remain the same, because their definitions are quite simple, but for energy and motion, the definitions are quite different.

2.3 time

There is only one thing to mention: in Galileo relativity, time is absolute, so velocity and acceleration depend on change in position vector; the only difference between different reference frames is different representation of the position vector. But in special relativity, time changes.

2.4 Describe motion in different reference frames

2.4.1 Concepts

Different inertial reference frames give different descriptions of the same event. The position vector and velocity vector depend on the choice of the reference frame, but in Galileo relativity time is absolute, and acceleration is the same in all inertial frames.

In the standard configuration, S' moves along the $+x$ direction with constant velocity $\vec{V}$ relative to S . It does not matter which frame is called S and which is called S' , as long as we are consistent.

Let frame S' move with constant velocity $\vec{V}$ relative to frame S . If the origins coincide at $t = 0$, then the position vectors are related by

$$\vec{r} = \vec{r}' + \vec{V}t, \quad \vec{r}' = \vec{r} - \vec{V}t.$$

Differentiating with respect to time gives the velocity transformation

$$\vec{v}' = \vec{v} - \vec{V}, \quad \vec{v} = \vec{v}' + \vec{V}.$$

Since $\vec{V}$ is constant,

$$\vec{a}' = \vec{a}.$$

In Galileo relativity time is absolute.

$$t' = t,$$

In the standard configuration, where S' moves along the positive x direction,

$$x' = x - Vt, \quad y' = y, \quad z' = z, \quad t' = t,$$

and

$$v'_x = v_x - V, \quad v'_y = v_y, \quad v'_z = v_z.$$

In Galileo relativity: time is absolute; position and velocity depend on the reference frame; acceleration is universal in all inertial frames.

The full derivation is given again in Section 3.1.

2.4.2 Original note

We define the position vector $\vec{r}$ and the velocity vector $\vec{v}$. It depends on the choice of the reference frame.

If we choose different reference frames, the position vector and the velocity vector will be different, related by the following equation*s:

$$\vec{r} = \vec{r}' + \vec{R}$$

(we assume that the position vector of particle X in frame S is $\vec{r}$ and in frame S' is $\vec{r}'$, and the position vector of the origin of frame S' relative to frame S is $\vec{R}$)

Use differentiation, we can get the relation of the velocity vector in different reference frames; overall it is similar to the position vector:

$$\begin{aligned}\vec{v}' &= \frac{d\vec{r}'}{dt} \\ &= \frac{d\vec{r}}{dt} - \frac{d\vec{R}}{dt} \\ &= \vec{v} - \vec{V} \\ \vec{v} &= \vec{v}' + \vec{V} \\ &\text{(inverse relation)}\end{aligned}$$

Remember the time in Galileo relativity is absolute, $t' = t$, which allows us to build this relationship. The relation of the position vector is similar to the relation of velocity, as said, “velocity and acceleration depend on change in position vector”.

3 Galilean relativity and the Michelson-Morley experiment

Relativity is the relationship of one reference frame to another.

3.1 Review Galileo transformation

3.1.1 Relations

Let frame S' move with constant velocity $\vec{V}$ relative to frame S . For a particle, denote its position by $\vec{r}$ in S and $\vec{r}'$ in S' . If the origins coincide at $t = 0$, then

$$\vec{R} = \vec{V}t,$$

and

$$\vec{r} = \vec{r}' + \vec{R} = \vec{r}' + \vec{V}t.$$

Therefore

$$\vec{r}' = \vec{r} - \vec{V}t.$$

Differentiating with respect to time,

$$\vec{v}' = \frac{d\vec{r}'}{dt} = \vec{v} - \vec{V},$$

or

$$\vec{v}' = \vec{v} - \vec{V}.$$

The inverse transformation is

$$\vec{v} = \vec{v}' + \vec{V}.$$

Since $\vec{V}$ is constant,

$$\frac{d\vec{V}}{dt} = 0,$$

so

$$\boxed{\vec{a}' = \vec{a}.}$$

If S' is non-inertial, i.e. $\vec{V} = \vec{V}(t)$, then

$$\vec{a}' = \vec{a} - \vec{A}, \quad \vec{A} = \frac{d\vec{V}}{dt}.$$

And at last

$$\boxed{\text{In Galileo relativity } t' = t, \text{ time is absolute.}}$$

3.1.2 Original Note and Example

A frame S moves with a constant velocity $\vec{V}$ relative to another frame S' . In sight of Galileo relativity,

$$\begin{aligned} \vec{R} &= \vec{V}t \\ \vec{r}' &= \vec{r} - \vec{R} \\ \dot{\vec{R}} &= \vec{V} = \text{const} \\ \vec{r}' &= \vec{r} - \vec{R} = \vec{r} - \vec{V}t \end{aligned}$$

the position transformation does not transform time.

$$\begin{aligned} t' &= t \\ \vec{u}' &= \vec{u} - \vec{V} \\ \vec{a}' &= \vec{a} - \vec{A} \end{aligned}$$

(where $\vec{a}'$ is the acceleration in frame S' , $\vec{a}$ is the acceleration in frame S , and $\vec{A} = d\vec{V}/dt$ is the acceleration of frame S' relative to frame S). In inertial frames, $\vec{A} = 0$, so $\vec{a}' = \vec{a}$.

$$\boxed{\text{Acceleration in Galileo relativity is universal.}}$$

example, ship constant velocity, a ball dropped from the crow's nest will fall straight down relative to the ship, even though the ship is moving relative to the ground.

3.2 Introduction to Michelson–Morley experiment

- beginning of the 1800s, Young interference experiment— wave nature of light.
- wave requires a medium for propagation, leading to the hypothesis of the luminiferous aether.

Galileo relativity works well for mechanics, but it is not compatible with Maxwell's electromagnetism. Maxwell's equation*s give a fixed speed of light c , and experiments always measured the same speed of light, independent of the motion of the Earth.

At that time, physicists assumed light needed a medium called the luminiferous aether. The Michelson–Morley experiment was designed to detect the Earth's motion relative to this aether.

3.2.1 Hand Clapping Experiment

In this experiment, two people stand a certain distance apart and clap their hands when they hear another's clap.

- case1 platform is stationary relative to the ground.
- case2 platform is moving relative to the ground.
- case3 is perpendicular to the direction of motion.

for case 1

$$t = \frac{2L}{v_{\text{sound}}}$$

for case 2

$$t = \frac{2Lv_{\text{sound}}}{v_{\text{sound}}^2 - v_{\text{platform}}^2}$$

for case 3

$$t = \frac{2L}{\sqrt{v_{\text{sound}}^2 - v_{\text{platform}}^2}}$$

Clearly,

$$t_2 \geq t_3$$

As a scientific check, when $v_{\text{platform}} = 0$, we recover

$$t = \frac{2L}{v_{\text{sound}}}.$$

3.2.2 Michelson-Morley experiment

The main idea is not to measure T_1 and T_2 individually, we have to use a beam splitter.

We look at the interference pattern. If the two travel times are different, Because the waves are out of phase, if the time difference exists, the interference pattern will shift.

And if no shift takeplace, it means the two travel times are equal. No aether was detected.

$$T_1 = t_f + t_b = \frac{2Lc}{c^2 - v^2} \quad (\text{parallel arm})$$

$$T_2 = \frac{2L}{\sqrt{c^2 - v^2}} \quad (\text{perpendicular arm})$$

$$T_2 = T_1 \sqrt{1 - \frac{v^2}{c^2}}$$

Lorentz and FitzGerald proposed that objects contract along the direction of motion, with the Lorentz factor, in order to explain the null result.

$$\sqrt{1 - \frac{v^2}{c^2}}$$

This was mathematically close to the correct transformation, but for the wrong physical reason. Einstein later explained the null result by postulating that the speed of light is the same in all inertial frames.